Viva and Conceptual Q&A

Distance Vector Routing with RIP version 2 in Cisco Packet Tracer

Your project at a glance

One-line summary: three routers connect three LANs over two serial links. There are no static routes; each router learns the other networks from its neighbours using RIP version 2.

Device	Interface	IP address	Network	Note
Router0	Fa0/0	192.168.1.1	LAN 1: 192.168.1.0/24	Gateway for PC0, PC1
Router0	S2/0	10.0.0.1	10.0.0.0/8	To Router1 (DTE)
Router1	Fa0/0	192.168.2.1	LAN 2: 192.168.2.0/24	Gateway for PC2, PC3
Router1	S2/0	10.0.0.2	10.0.0.0/8	To Router0 (DCE, sets the clock)
Router1	S3/0	11.0.0.1	11.0.0.0/8	To Router2 (DTE)
Router2	Fa0/0	192.168.3.1	LAN 3: 192.168.3.0/24	Gateway for PC4, PC5
Router2	S2/0	11.0.0.2	11.0.0.0/8	To Router1 (DCE, sets the clock)

PCs: each LAN has .2 and .3 (PC0 192.168.1.2, PC1 192.168.1.3, PC2 192.168.2.2, PC3 192.168.2.3, PC4 192.168.3.2, PC5 192.168.3.3).

Router0 routing table (seen in your Packet Tracer run):

```
C 10.0.0.0/8 is directly connected, Serial2/0
R 11.0.0.0/8 [120/1] via 10.0.0.2, 00:00:08, Serial2/0
C 192.168.1.0/24 is directly connected, FastEthernet0/0
R 192.168.2.0/24 [120/1] via 10.0.0.2, 00:00:08, Serial2/0
R 192.168.3.0/24 [120/2] via 10.0.0.2, 00:00:08, Serial2/0
```

Item	RIP value	Item	RIP value
Metric	Hop count	Update timer	30 seconds
Maximum hops	15 (16 = unreachable)	Invalid timer	180 seconds
Administrative distance	120	Hold-down timer	180 seconds
Transport	UDP port 520	Flush timer	240 seconds
RIPv2 destination	Multicast 224.0.0.9	Algorithm	Bellman-Ford

Viva tip: answer in one or two sentences, then point at your network. "Router0 reaches LAN 3 at 2 hops through Router1" is stronger than a textbook definition.

1. About your project

Q1. What is your project about?

It demonstrates distance vector routing. Three routers running RIP version 2 connect three LANs, and each router learns the remote networks automatically, with no static routes.

Q2. What devices did you use?

Three Router-PT routers, three Switch-PT switches and six PC-PT computers, with two PCs in each LAN. Routers are joined by serial links; everything else uses copper straight-through cables.

Q3. Why did you choose RIP?

RIP is the classic distance vector protocol. It is simple to configure and its hop-count metric makes the Bellman-Ford idea easy to see in a routing table.

Q4. Why RIP version 2 and not version 1?

RIPv2 is classless: it sends the subnet mask with each route, supports VLSM and authentication, and multicasts updates to 224.0.0.9 instead of broadcasting them.

Q5. Why is there a separate network between each pair of routers?

Each router interface must be in a different network. The serial links are their own networks, 10.0.0.0/8 and 11.0.0.0/8, and routers forward traffic between networks.

Q6. Is /8 a good choice for a link between two routers?

Not in a real network. A point-to-point link needs only two addresses, so /30 is the efficient choice. A /8 wastes millions of addresses, but it keeps this lab simple.

Q7. Is 11.0.0.0/8 a private network?

No. 10.0.0.0/8 is private, but 11.0.0.0/8 is public address space. It is harmless in an isolated lab, but a real network should use private ranges such as 10.0.0.0/8, 172.16.0.0/12 or 192.168.0.0/16.

Q8. What is the role of the switches?

They connect the PCs and the router within a LAN, forwarding frames by MAC address at Layer 2. They do not route and are not counted as hops.

Q9. What is the default gateway of PC4, and why is it needed?

It is 192.168.3.1, Router2's LAN interface. A PC sends any traffic for another network to its gateway, because the PC itself cannot reach that network directly.

Q10. Which cables did you use, and why?

Straight-through copper between unlike devices (PC to switch, switch to router) and serial cables between routers. Crossover cables are traditionally used between like devices.

Q11. What are DCE and DTE on your serial links?

The DCE end provides the clock, and the DTE end follows it. Router1 S2/0 and Router2 S2/0 are the DCE ends, so they have the clock rate command. Without a clock, the link stays down.

Q12. What does the address 10.0.0.2 in Router0's table mean?

It is the next hop: Router1's interface on the link to Router0. That is the "vector", the direction Router0 sends the packets.

Q13. What are the weaknesses of your topology?

It is a line with no redundant links. If Router1 fails, LAN 1 and LAN 3 lose contact with each other, and RIP cannot reroute because no other path exists. Adding a Router0 to Router2 link would fix that.

Q14. How would you improve this project?

Add a redundant link, use /30 on the router links, use private addresses throughout and enable RIPv2 authentication. For a larger network, I would move to OSPF.

2. Distance vector concepts

Q15. What is routing?

Routing is choosing the path a packet takes between networks. A router looks up the destination network in its routing table and forwards the packet towards the next hop.

Q16. What is the difference between static and dynamic routing?

An administrator types static routes by hand, and they do not adapt on their own. Dynamic routing protocols learn routes automatically and update them when the network changes, at the cost of some bandwidth and CPU.

Q17. What is distance vector routing?

A dynamic routing method where each router knows only its neighbours. Routers share their routing tables, and each one picks the path with the lowest cost.

Q18. What do "distance" and "vector" mean?

Distance is how far away a network is, measured by a metric. In RIP that is the hop count. Vector is the direction: the next-hop router or exit interface.

Q19. Why is it called "routing by rumour"?

A router never sees the whole network. It trusts whatever its neighbours report, even if that information is out of date.

Q20. State the Bellman-Ford equation.

$D_x(y) = \min_v \{ c(x, v) + D_v(y) \}$. The best distance from x to y is the lowest value, over all neighbours v, of the link cost to v plus the distance v reports to y.

Q21. Apply Bellman-Ford to your network.

For Router0 to LAN 3, the only neighbour is Router1. The link cost is 1 hop, and Router1 reports LAN 3 at 1 hop, so the distance is 1 + 1 = 2 hops via Router1. This matches [120/2] in the table.

Q22. How does a router build its table, step by step?

It starts with its connected networks at distance 0. It receives its neighbours' tables, adds the link cost to each distance, and keeps a route if it is new or better. This repeats until nothing changes.

Q23. What is convergence?

Convergence is the state where all routers agree on the network and their tables stop changing. Faster convergence means less time with wrong or missing routes.

Q24. How many update rounds does your network need to converge?

Two rounds. News travels one hop per round: Router2 tells Router1 about LAN 3 in round one, then Router1 tells Router0 in round two.

Q25. What is the difference between a routing table and a forwarding decision?

The table holds known networks and their next hops. For each packet, the router picks the matching entry with the longest prefix and forwards the packet out of that entry's interface.

Q26. What is the difference between an IGP and an EGP?

An interior gateway protocol such as RIP or OSPF routes inside one organisation's network, or autonomous system. An exterior protocol, BGP, routes between autonomous systems.

3. RIP

Q27. What is the metric in RIP?

Hop count, the number of routers a packet must cross to reach the network. A directly connected network has metric 0.

Q28. What is the maximum hop count, and why?

The maximum is 15, and 16 means unreachable. The limit stops count-to-infinity from going on forever, but it also limits RIP to small networks.

Q29. What is the main weakness of using hop count?

It ignores bandwidth. RIP prefers 2 hops over slow links to 3 hops over very fast links.

Q30. What does [120/2] mean?

120 is the administrative distance of RIP, and 2 is the metric, or hop count.

Q31. What is administrative distance?

It ranks how much a router trusts a source when several protocols offer the same network, and lower wins. Connected is 0, static is 1, EIGRP is 90, OSPF is 110 and RIP is 120.

Q32. Which protocol and port does RIP use?

RIP messages travel in UDP on port 520.

Q33. How often does RIP send updates?

It sends its full routing table every 30 seconds. A triggered update goes out immediately when a route changes.

Q34. Explain the RIP timers.

Update (30 s) sets how often tables are sent. Invalid (180 s) marks a route as possibly down when no updates arrive. Hold-down (180 s) ignores worse news while a route settles. Flush (240 s) removes the route.

Q35. Compare RIPv1 and RIPv2.

RIPv1 is classful, broadcasts updates and has no authentication. RIPv2 is classless, sends subnet masks, multicasts to 224.0.0.9 and supports authentication. Both use hop count with a limit of 15.

Q36. What is the difference between classful and classless routing?

Classful routing assumes the mask from the address class, so updates carry no mask. Classless routing sends the mask with each route, which allows VLSM and CIDR.

Q37. What does "no auto-summary" do?

It stops RIP from summarising subnets to their classful boundary, so exact subnets are advertised. Your networks already match classful boundaries, so routing would also work without it, but it is the correct practice for RIPv2.

Q38. What does the "network" command do under router rip?

It is classful and selects interfaces. Interfaces in that network send and receive RIP, and their networks are advertised. It does not mean "advertise this one route only".

Q39. Why does Router1 have three network statements while the others have two?

Router1 has three connected networks: LAN 2, 10.0.0.0 and 11.0.0.0. Router0 and Router2 each have one LAN and one serial link.

Q40. What does passive-interface do, and why did you use it on Fa0/0?

It stops RIP updates going out of that interface, while the LAN is still advertised to other routers. PCs do not run RIP, so updates on the LAN would be wasted traffic.

Q41. Does RIP support load balancing?

Yes, across equal-cost paths. Your network has only one path to each destination, so it does not use it.

4. Routing loops and count-to-infinity

Q42. What is a routing loop?

Packets circle between routers because their tables point at each other. The TTL eventually drops the packet, but bandwidth is wasted until then.

Q43. What is the count-to-infinity problem?

After a network fails, routers keep trusting each other's stale routes, and the metric grows with each exchange until it reaches infinity. In RIP, infinity is 16.

Q44. Show count-to-infinity on your network.

LAN 3 fails at Router2. Without protection, Router1 still advertises LAN 3 at 1 hop, so Router2 thinks it can reach LAN 3 at 2 hops through Router1. Router1 then updates to 3 hops through Router2, and the count rises to 16.

Q45. What is split horizon?

A router never advertises a route back out of the interface it learned it on. Router1 learned about LAN 3 from Router2, so it does not advertise LAN 3 back to Router2.

Q46. What is route poisoning?

When a network fails, the router immediately advertises it with metric 16, telling everyone it is unreachable, rather than waiting for the route to time out.

Q47. What is poison reverse?

When a router hears a poisoned route, it sends the poison back out of the same interface, confirming the news. It deliberately overrides split horizon for that route.

Q48. What are hold-down timers?

After a route goes down, the router ignores worse or equal news about it for a while (180 s by default), so stale information cannot bring the route back.

Q49. What are triggered updates?

A router sends an update as soon as a route changes, rather than waiting for the 30-second timer, which speeds up convergence.

Q50. Does split horizon solve every loop?

No. It stops loops between two neighbours, but loops through three or more routers can still form in meshed networks. That is why it is combined with hold-down timers and the hop limit.

5. Distance vector vs link state

	Distance vector	Link state
A router knows	Only what its neighbours report	A full map of the network
Algorithm	Bellman-Ford	Dijkstra (shortest path first)
Updates	Whole table, periodically	Only changes, when they happen
Convergence	Slow, with loop risk	Fast
CPU and memory	Low	Higher
Metric example	RIP: hop count	OSPF: cost based on bandwidth
Best for	Small, simple networks	Large networks
Examples	RIP, IGRP	OSPF, IS-IS

Q51. Why do large networks prefer OSPF over RIP?

OSPF has no 15-hop limit, converges quickly, uses a bandwidth-based metric, and sends only changes rather than whole tables.

Q52. When would RIP still be a reasonable choice?

In a small, simple network where easy setup matters more than fast convergence, or for teaching routing concepts.

Q53. Is EIGRP distance vector?

EIGRP is usually called an advanced distance vector, or hybrid, protocol. It uses the DUAL algorithm and sends partial updates rather than full tables.

Q54. What type of protocol is BGP?

BGP is a path vector protocol. It tracks the list of autonomous systems a route passes through, and it is used between autonomous systems on the internet.

6. How a packet travels: PC0 to PC4

Q55. Walk through a ping from PC0 to PC4.

PC0 sees that 192.168.3.2 is on another network, so it sends the packet to its gateway, 192.168.1.1. Router0 forwards it to 10.0.0.2 using its RIP route, Router1 forwards it to 11.0.0.2, and Router2 delivers it to PC4 on LAN 3.

Q56. What changes at each hop, and what stays the same?

The source and destination IP addresses stay the same. The Layer 2 header is rebuilt on each link, and each router decrements the TTL.

Q57. Why did the first ping time out?

The first packet is lost while ARP resolves MAC addresses on the Ethernet segments. After that the addresses are cached, so the following pings succeed.

Q58. Why was the TTL in the replies 125?

PC4 sends the reply with a TTL of 128, and each of the three routers subtracts 1, so 128 - 3 = 125. This proves the packet crossed all three routers.

Q59. Is ARP used on the serial links?

Not in your network. ARP resolves IP addresses to MAC addresses on Ethernet. The serial links are point-to-point with the Cisco default encapsulation, HDLC, so there is only one possible receiver.

Q60. Which OSI layers are involved?

Switches work at Layer 2 with MAC addresses, and routers work at Layer 3 with IP addresses. RIP messages travel in UDP, so they use Layer 4 transport, though RIP's job is network-layer routing.

7. Configuration and commands

Q61. Show the RIP configuration on Router0.

The commands are below. Router1 and Router2 are the same, with their own networks.

```
router rip
 version 2
 passive-interface FastEthernet0/0
 network 10.0.0.0
 network 192.168.1.0
 no auto-summary
```

Command	What it shows or does
`show ip route`	Routing table with C and R routes
`show ip protocols`	RIP version, timers, networks and passive interfaces
`show ip interface brief`	Each interface's IP address and up/down status
`show controllers serial 2/0`	Whether the serial end is DCE or DTE, and its clock
`debug ip rip`	RIP updates live; stop it with undebug all
`show running-config`	The current configuration in RAM
`copy running-config startup-config`	Saves the configuration to NVRAM
`ping / tracert`	Tests reachability and shows each hop (run on the PC)

Q62. What is the difference between running-config and startup-config?

Running-config is the active configuration in RAM and is lost at power-off. Startup-config is saved in NVRAM and loads at boot. Your routers have theirs saved.

Q63. What does "no ip domain-lookup" do?

It stops the router treating a mistyped command as a hostname and trying a DNS lookup, so a typo does not freeze the CLI for several seconds.

Q64. What does "spanning-tree portfast" do on your switches?

It moves a port straight to forwarding, skipping the roughly 30-second listening and learning wait. It is safe here because the switches have no loops between them.

Q65. What does "no shutdown" do?

It enables an interface. Router interfaces are shut down by default, so each one in use needs this command.

8. Demo and troubleshooting

Q66. What is the difference between Realtime and Simulation mode?

Realtime runs the network continuously, as it would really work. Simulation pauses time so you can step through events and open each packet to inspect its headers.

Q67. What are your scenarios for?

They are saved ping tests: same LAN, LAN 1 to LAN 2, LAN 2 to LAN 3, LAN 1 to LAN 3, and all LANs together. Together they test every pair of routers.

Q68. A PDU shows Failed the first time, then Successful. Why?

The first attempt can run out of time during ARP, or fire before RIP has converged. Once addresses and routes are in place, it succeeds.

Q69. What happens if you shut down Router2's Fa0/0?

LAN 3 goes down. Router2 poisons the route with metric 16, Router1 and Router0 learn it is unreachable, and pings from PC0 to PC4 fail. After no shutdown, the route returns. *Try this once before your viva.*

Q70. An R route is missing from a router. What do you check?

Check that the interfaces are up with show ip interface brief, then check version and network statements with show ip protocols, then watch updates with debug ip rip. Also allow time for convergence.

Q71. A serial link is down. What do you check?

Check that both ends have no shutdown and that addresses are in the same network. Then check that the DCE end has a clock rate, using show controllers.

Q72. PCs in the same LAN can ping each other, but not other LANs. What is wrong?

The likely causes are a wrong or missing default gateway on the PC, the router's LAN interface being down, or RIP not advertising the route.

Q73. What would happen if PC0's gateway were wrong?

PC0 could still reach PC1 in its own LAN, but traffic to other LANs would fail because it could not reach a working router.

Q74. If RIPv2 were on Router0 and RIPv1 on Router1, what would happen?

Router1 in version 1 mode ignores version 2 updates, and Router0 ignores version 1 updates, so routes would not be exchanged. All routers need the same version, or explicit send and receive settings.

9. Rapid-fire answers

Question	Answer	Question	Answer
RIP full form	Routing Information Protocol	RIP multicast address	224.0.0.9
Algorithm	Bellman-Ford	RIP port	UDP 520
Metric	Hop count	Administrative distance	120
Maximum hops	15	Unreachable metric	16
Update interval	30 seconds	Flush timer	240 seconds
RIPv2 standard	RFC 2453	RIPv1 standard	RFC 1058
OSPF algorithm	Dijkstra	OSPF administrative distance	110
Routers crossed, PC0 to PC4	3	RIP metric, Router0 to LAN 3	2
Layer of a router	Layer 3	Layer of a switch	Layer 2
Clock rate is set on	DCE end	Your DCE ends	Router1 S2/0, Router2 S2/0

Hops vs routers: PC0 to PC4 crosses 3 routers (so TTL drops by 3), but Router0's metric to LAN 3 is 2, because it counts the routers after Router0.